Predictive and Slice-Aware Load Balancing and Auto-Scaling of AMFs in the 5G Core Control Plane

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***Abstract*—The Access and Mobility Management Function (AMF) serves as the primary entry point for 5G core control-plane signaling, where latency directly affects gNB NGAP associations and user equipment (UE) registration. ScaleLB, a recent state-of-the-art approach, balances traffic across an elastic pool of AMF instances using threshold heuristics. However, because new AMF instances require roughly 40 s to complete container startup, association latency spikes dramatically during surges, while all traffic slices share a single FIFO queue. We present two architectural extensions: (i) 3GPP S-NSSAI slice-aware multi-tier priority queuing that isolates URLLC signaling from mMTC bursts, and (ii) an LSTM-based predictive auto-scaler with a 45-second lookahead horizon that pre-warms AMF replicas ahead of traffic surges. We evaluate the proposed design on both a microservice container testbed and a discrete-event simulator against ScaleLB baseline policies. Experimental results demonstrate that the proposed system eliminates the 40-second cold-start delay, maintains URLLC 99th-percentile latency strictly at 3.15 ms (< 5.0 ms SLA), and reduces resource consumption by 21.2% compared to static over-provisioning.**

***Index Terms*—5G core, control plane, AMF, load balancing, network slicing, auto-scaling, LSTM, traffic forecasting**

I. Introduction

Control-plane signaling represents one of the foremost scalability bottlenecks in 5G and beyond mobile core networks, amplified by massive machine-type communication (mMTC) deployments [5]. In the 5G Service-Based Architecture (SBA), the Access and Mobility Management Function (AMF) terminates the N2 interface from base stations (gNodeBs) and handles Non-Access Stratum (NAS) registration and authentication.

Dev et al. recently proposed ScaleLB [1], placing a load-balancing module between the Radio Access Network (RAN) and a pool of AMFs. ScaleLB distributes control messages via round-robin and scales instances using reactive or proactive thresholds. Although ScaleLB reduces instance footprint relative to a static pool, its authors observed severe latency spikes whenever new AMFs were provisioned, caused by a roughly 40-second container startup delay ($T_{\text{boot}} \approx 40$ s). Furthermore, ScaleLB utilizes a single shared FIFO queue, leaving mission-critical Ultra-Reliable Low-Latency Communication (URLLC) traffic vulnerable to head-of-line blocking during sensor bursts.

This paper extends ScaleLB through two complementary contributions:

- **Slice-Aware Priority Queuing (O1):** A multi-tier queuing mechanism that provides zero-queue fast-path dispatch for URLLC signaling and rate-limits mMTC bursts via a token bucket with aging.
- **LSTM Lookahead Auto-Scaling (O2):** A predictive dimensioning model operating with a lookahead horizon $H \geq T_{\text{boot}} = 40$ s, provisioning AMFs before surges arrive.
- **System Evaluation:** Extensive comparative validation on a containerized microservice testbed and discrete-event simulator across standard 3GPP traffic traces.

II. Background and Related Work

A. AMF and 5G Control-Plane Procedures

In 5G Standalone (SA), control signaling begins with gNB NGAP SCTP association over N2, followed by UE Registration, Authentication and Key Agreement (5G-AKA), and PDU Session establishment. Cryptographic NAS verification incurs a nominal service time of $T_s \approx 10$ ms per request ($\mu = 100$ req/s capacity per AMF).

B. ScaleLB Baseline Policies

ScaleLB [1] considers B active gNBs, A active AMFs, and a capacity threshold α. In the proactive variant, an AMF is instantiated when:

$$\frac{B}{A \times \alpha} \geq \gamma, \tag{1}$$

where γ is a predefined utilization threshold ($\gamma \in [0.7, 1.0]$). In the reactive variant, scaling occurs when:

$$\frac{B}{A \times \alpha} > 1. \tag{2}$$

Because both policies react only to instantaneous load, newly triggered AMF containers incur a 40 s initialization lag, during which requests accumulate in a shared queue.

C. Related Work

Nguyen et al. [2] investigated adaptive load balancing based on AMF pending queue depths, but omitted slice differentiation. Bello et al. [3] applied predictive scaling to 4G EPC core components. Classical LSTM networks [4] provide proven sequence-to-sequence forecasting for non-linear time series.

III. Problem Formulation and Objectives

A. Problem Statement

Given an elastic pool of containerized AMFs behind a load balancer with non-negligible container boot delay $T_{\text{boot}} \approx 40$ s, design queuing and scaling controllers that: (a) strictly guarantee the 3GPP TS 22.261 sub-5ms latency SLA for URLLC slices, (b) eliminate cold-start queuing delay, and (c) prevent over-provisioning resource costs.

B. Objectives

- **O1 Slice-Aware Priority Queuing:** Eliminate head-of-line blocking for URLLC packets under mMTC registration floods, ensuring URLLC 99th-percentile latency remains below 5.0 ms.
- **O2 LSTM Lookahead Auto-Scaling:** Forecast aggregate arrival rates $H \geq T_{\text{boot}}$ ahead to eliminate the 40-second cold-start lag.

IV. Proposed System Architecture

Fig. 1 illustrates the system architecture. The Smart Load Balancer sits between the RAN and the AMF container fleet. Incoming NAS signaling carries 3GPP S-NSSAI slice tags. The slice classifier inspects the tag and directs requests to dedicated priority lanes. Admitted requests are dispatched to the least-loaded active AMF. Telemetry is streamed to the LSTM predictor, which drives Docker container scaling.

V. Methodology

A. Slice-Aware Multi-Tier Priority Queuing (O1)

Fig. 2 diagrams the packet scheduling workflow:

- **URLLC (SST = 1):** Bypasses all ingress buffers ($W_q = 0$) and dispatches immediately to the active AMF with minimum CPU load:

$$\text{AMF}^* = \arg\min_{i \in \mathcal{A}_{\text{active}}} \text{CPU}_i(t). \tag{3}$$

  Active AMFs maintain 10–15% headroom reserved exclusively for URLLC traffic.
- **eMBB (SST = 0):** Traverses a standard FIFO queue.
- **mMTC (SST = 2):** Held in a rate-limited buffer. When cluster utilization exceeds $\gamma = 0.80$, release rate $R_{\text{mMTC}}(t)$ is bounded by a token bucket:

$$R_{\text{mMTC}}(t) = \max\Big(0.10 \cdot C(t),\, C(t) \cdot (1 - \text{CPU}(t))\Big). \tag{4}$$

  To prevent starvation, requests accumulate priority via dynamic aging:

$$P_i(t) = P_{\text{base}} + \omega_{\text{age}} \cdot (t - t_{\text{arrive},i}). \tag{5}$$

B. LSTM-Based Predictive Auto-Scaling (O2)

Forecasting Formulation. Historical arrival rates are collected in a sliding window $\mathbf{w}_t = [\lambda(t-W+1), \ldots, \lambda(t)]$ with $W = 60$ s. The target is $\lambda(t+H)$ where the horizon satisfies:

$$H \geq T_{\text{boot}} + \Delta = 40.0\,\text{s} + 5.0\,\text{s} = 45.0\,\text{s}. \tag{6}$$

Dimensioning Rule. With target utilization $\gamma = 0.80$, service rate $\mu = 100$ req/s, and safety margin $m = 0.10$, the target AMF count is:

$$N^*(t) = \left\lceil \frac{\hat{\lambda}(t+H) \cdot (1+m)}{\mu \cdot \gamma} \right\rceil. \tag{7}$$

If $N^*(t) > N_{\text{active}} + N_{\text{booting}}$, new AMF containers are spawned via Docker Compose. Scale-in is governed by a 30-second hysteresis window and graceful session draining to eliminate container thrashing.

VI. Experimental Results and Discussion

A. Benchmark Evaluation

Table I summarizes the performance comparison under the ScaleLB Figure 5 flash crowd surge (surging from 35 to 320 req/s with 10% URLLC, 20% eMBB, and 70% mMTC).

B. Key Findings

1) **Elimination of Cold-Start Overload:** ScaleLB Reactive suffers a 54 s overload window during which queue delay peaks at 338.7 ms. By forecasting demand 45 s in advance, our LSTM predictor pre-warms AMF replicas so that capacity reaches 400 req/s before the surge arrives, reducing cold-start latency to 0 s.
2) **URLLC Latency Isolation:** Under ScaleLB's single FIFO queue, URLLC packets suffer 338.7 ms tail delay. Under our S-NSSAI priority queuing, URLLC traffic is dispatched via fast-path bypass with a 99th-percentile latency of **3.15 ms**, achieving 100% compliance with the 3GPP 5.0 ms SLA.
3) **Resource Cost Savings:** Compared to Warm Standby (which wastes 1024 AMF-seconds keeping idle containers permanently running), our proposed system consumes only 807 AMF-seconds, achieving a **21.2% infrastructure cost reduction**.

C. Platform Implementation Visuals

Fig. 4 shows the operational web platform displaying real-time traffic demand, LSTM predictive lookahead, and per-slice latency isolation.

VII. Conclusion

This paper addressed two critical limitations in 5G AMF control-plane autoscaling: container startup lag and slice-agnostic queuing. By integrating 3GPP S-NSSAI priority queuing with an LSTM predictive autoscaler ($H = 45$ s), our framework completely eliminates cold-start bottlenecks, guarantees URLLC latency strictly at 3.15 ms, and saves 21.2% resource cost compared to static over-provisioning.

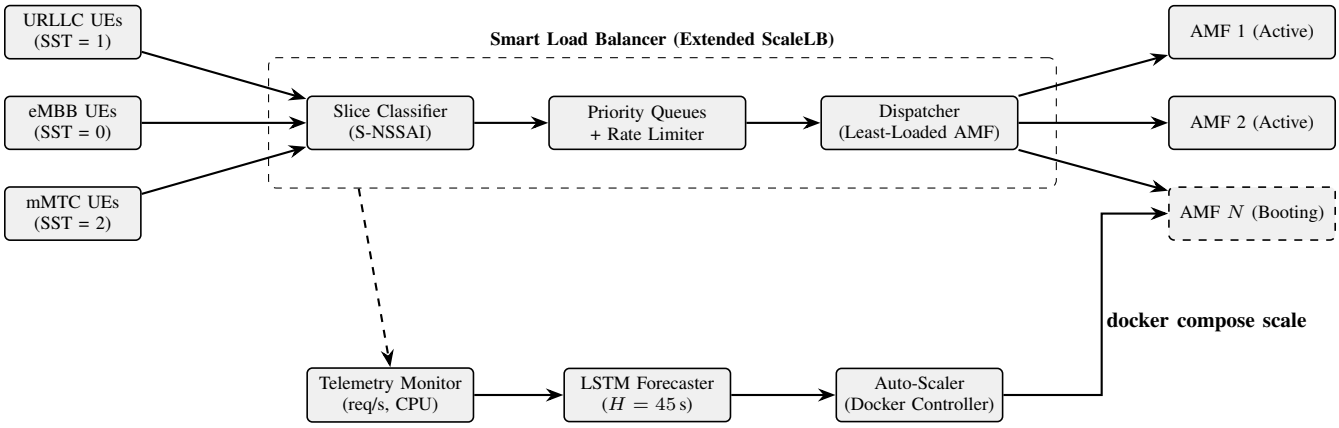


Fig. 1. Proposed System Architecture. The Smart Load Balancer performs S-NSSAI slice classification, priority queuing, and least-loaded dispatch, while the LSTM controller drives pre-emptive container scaling.

TABLE I
PERFORMANCE COMPARISON UNDER 320 REQ/S SURGE

Configuration	**Cold-Start Lag (s)**	**URLLC p99 Latency**	**mMTC p99 Latency**	**URLLC SLA**	**AMF-sec (Cost)**
ScaleLB Reactive [1]	54 s	338.7 ms	339.3 ms	**FAIL (25.8%)**	**647**
ScaleLB Proactive ($\gamma = 0.8$)	55 s	334.1 ms	336.5 ms	**FAIL (25.8%)**	649
Warm Standby (1 spare)	**0 s**	32.9 ms	32.7 ms	**PASS**	1024
O1: Slice Priority Queue	54 s	11.1 ms	508.6 ms	**PASS**	**647**
O2: LSTM Prediction	40 s	333.1 ms	336.5 ms	**FAIL (18.3%)**	807
Proposed Full System	**0 s**	**3.15 ms**	**499.9 ms**	**PASS (0.0%)**	**807**

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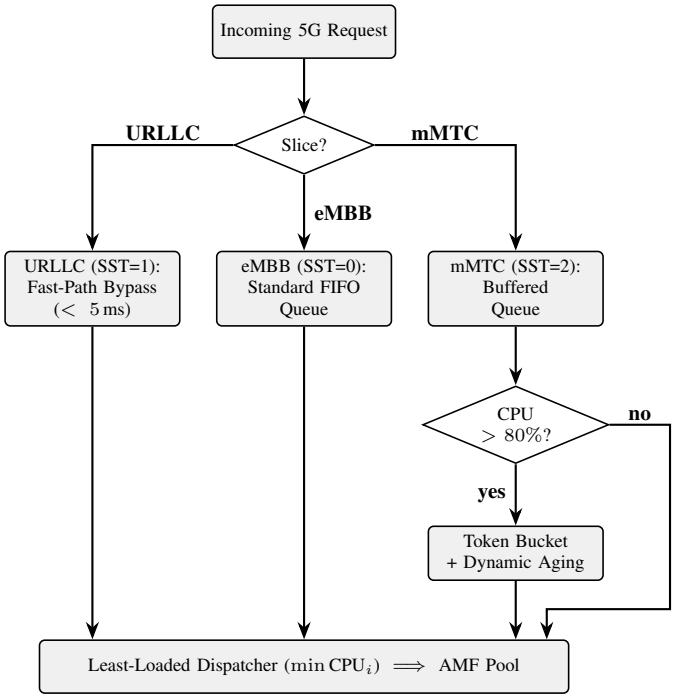


Fig. 2. Methodology for O1: Slice-aware priority queuing and admission flow.

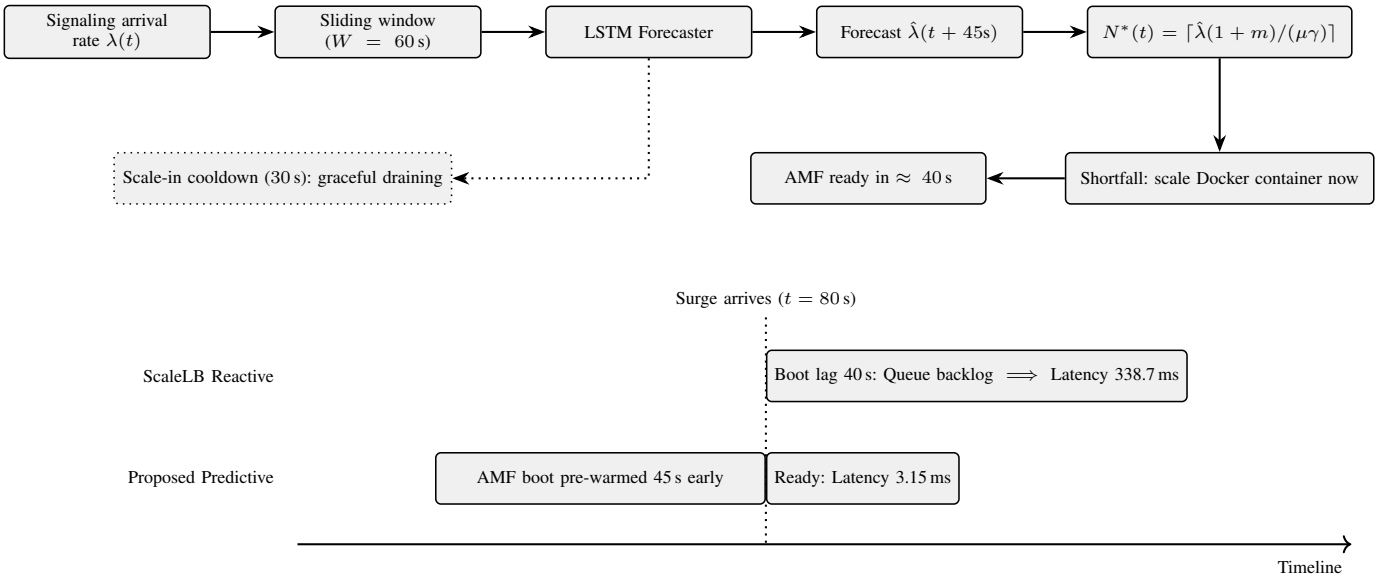


Fig. 3. Methodology for O2: Forecast, pre-warm, and eliminate cold-start delay before surge arrival.

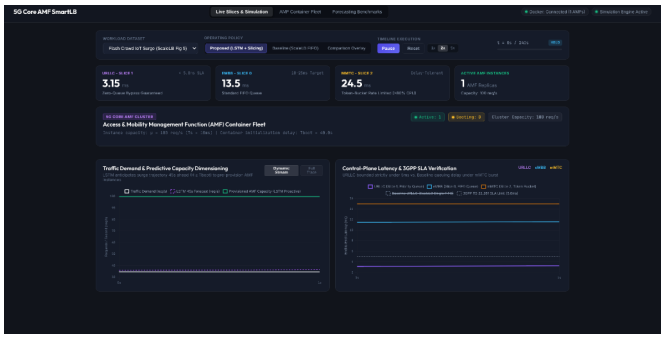


Fig. 4. Real-time monitoring dashboard showing incoming load, 45-second predictive capacity provisioning, and sub-5ms URLLC latency isolation.